

Adaptive Proximal Control for Federated Personalization

Supplementary Material

Anonymous submission

Experimental Protocol

Tables 1 and 2 give the protocol details behind the main-paper figures.

Controller Details

Adaptive proximal control replaces a fixed proximal coefficient with a bounded client-side control state. We refer to the FedProx instantiation as AdaFedProx and the Ditto instantiation as AdaDitto. For FedProx, this state is $\mu_i^{(t)}$; for Ditto, it is $\lambda_i^{(t)}$.

At the start of round t , the server broadcasts $w^{(t)}$ and the previous reference $L_g^{(t-1)}$. Client i evaluates the received global model on a local probe minibatch and obtains $\hat{L}_i^{(t)}$. The minibatch is taken from the shuffled local training loader before local optimization, using 10% of local training examples capped at 256 examples, with at least one example. The probe uses inference mode and no gradients. The client computes

$$g_i^{(t)} = \text{clip} \left(\log \frac{\hat{L}_i^{(t)} + \epsilon}{L_g^{(t-1)} + \epsilon}, 0, \tau \right).$$

For AdaDitto, the coefficient target is $\lambda_{\text{base}} + \alpha g_i^{(t)}$, then smoothed with rate γ and clipped to $[\lambda_{\min}, \lambda_{\max}]$. AdaFedProx uses the analogous μ notation. The resulting coefficient is used in the ordinary FedProx or Ditto proximal objective for that client’s local update. The client returns its model update and scalar probe loss, and the server updates the reference for future rounds using a median exponential moving average over the returned probe losses. The main paper lists the default AdaDitto values; the sensitivity scans below vary them around that default.

The controller does not change aggregation or the underlying FedProx and Ditto objectives beyond replacing a static coefficient with a controlled one. It preserves the model-update communication pattern and adds one scalar loss report per participating client and one scalar reference in the server broadcast.

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Controller Trajectories

Figure 1 shows that the adaptive coefficient changes smoothly rather than jumping to extreme values. The controller increases coupling under both CIFAR-pathological partial participation and Shakespeare, but settles in different ranges.

Controller Signal Validity

We audited whether the controller’s loss-gap signal tracks the anchoring need rather than merely producing a plausible coefficient trajectory. Table 3 reports post-warmup averages over participating clients: the clipped log loss-gap g_i , the fraction of positive gaps, and the learned final Ditto coefficient.

Within a run, the raw controller target is affine in the loss gap by construction, so the target-gap correlation checks implementation rather than mechanism. The cross-condition comparison is the relevant signal: on both CIFAR splits, 20% participation yields much larger gaps than full participation and correspondingly larger learned coefficients, matching the known shift toward $\lambda^* = 1.0$ under partial participation. This supports a mechanistic reading of the CIFAR shift result: the controller increases anchoring where the loss-gap signal says anchoring is needed.

The same audit also identifies a real limitation. The lower coefficient bound is load-bearing. When the known optimum is near zero, as in CIFAR-pathological full participation, AdaDitto cannot reach zero and slightly over-anchors. On Shakespeare, the loss gap is nearly zero, so AdaDitto’s parity with tuned Ditto appears to come mainly from the lower-bound floor rather than active gap-driven adaptation. This does not weaken the CIFAR participation-shift mechanism, but it scopes the controller: the gap is informative under CIFAR participation shift, while the floor acts as a safety net when the signal is weak.

Controller Sensitivity

We scanned 28 AdaDitto controller configurations on CIFAR-100 practical, varying gain, smoothing, EMA behavior, clipping threshold, and maximum coefficient. Final personalized accuracy ranged from 0.4762 to 0.4791, a spread of 0.29 points. The default configuration was tied-best (Table 4).

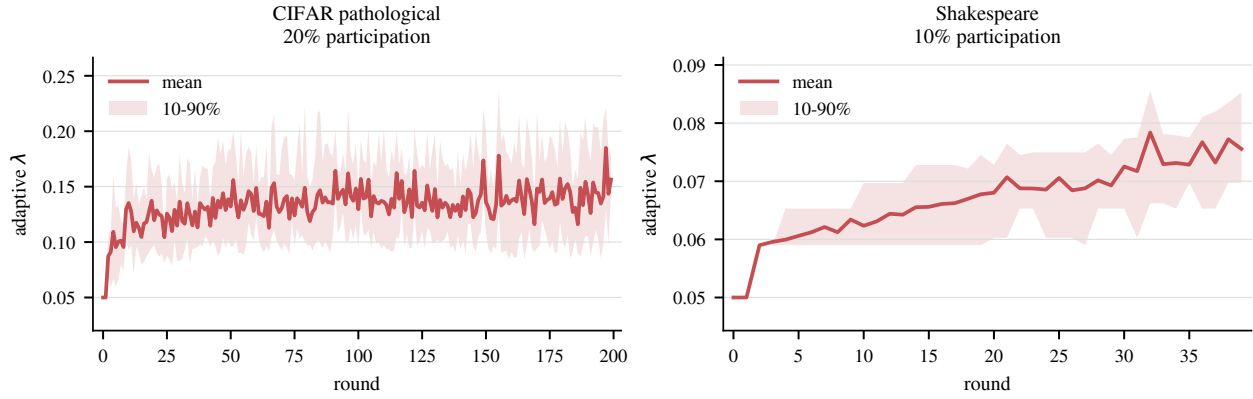


Figure 1: Adaptive Ditto coefficient trajectories. Lines show the mean coefficient over participating clients and seeds; bands show the 10th–90th percentile range.

We also ran an 11-configuration stress scan at the two headline conditions: CIFAR-100 pathological with 20% participation and Shakespeare with 10% participation. Table 5 reports the sorted-scan summary. On CIFAR-pathological, every controller configuration remains above the stale tune-once baseline and the default is median-ranked. On Shakespeare, the three-seed scan lies within 0.03 points and remains within 0.13 points of the tuned fixed-Ditto optimum. These scans test whether the headline results depend on a lucky default controller setting; they do not.

We also ablated controller mechanisms across both CIFAR splits. On the practical split, all modes were close relative to seed variation, so the random-adaptive row should not be read as a mechanism improvement. It is kept only as a sanity check: on the pathological split, principled modes clustered around 0.597, while random adaptive coefficients fell to 0.5863 (Table 6).

Tune-Once Regret Matrix

Table 7 reframes coefficient brittleness as deployment regret. Each target column has its own best fixed λ on the common CIFAR grid. Static rows select λ in the source condition and deploy that same value unchanged on each target.

A fixed λ is safe only when the deployment condition resembles the tuning condition. The $\lambda = 0$ setting selected by CIFAR-pathological under high participation is optimal there, but loses 4.7 points on pathological 20% participation and 7.2 points on practical 20% participation. AdaDitto does not match the per-condition oracle, but its observed regret is bounded by 3.2 points across the same targets. Thus the controller acts as a downside cap rather than an oracle replacement.

FedProx Negative Probe

FedProx’s proximal term did not improve CIFAR-100 in the tested regimes. This explains why AdaFedProx is mostly a parity result: when the best μ is approximately zero, there is no useful proximal signal for an adaptive μ controller to recover. Table 8 summarizes the negative probe.

Shakespeare Details

Shakespeare supplies the text-modality check. Its optimal direction differs from CIFAR-pathological: pure-local Ditto is poor, while nonzero coupling is essential. Table 9 shows that the best fixed coefficient remains stable across the tested participation levels, and Table 10 shows that AdaDitto matches tuned Ditto rather than producing a shift win.

Reproducibility Details

Tables 11 and 12 summarize implementation, data, hardware, randomness, reporting, and final training hyperparameters.

Dataset	Clients	Model	Partition	Participation	Rounds	Rep.	Role
CIFAR-100 practical	20	CNN	Dirichlet $\alpha = 0.1$, overlapping labels	full, 50%, 20%	200	mixed	fixed sweeps, core comparisons, controller scans
CIFAR-100 pathological	20	CNN	label shards, 10 labels/client	full, 50%, 20%	200	mixed	fixed sweeps, tune-once shift, core comparisons
Shakespeare	118	char-LSTM	LEAF speakers	10%, 50%	40	mixed	fixed sweeps, text-modality and controller checks

Table 1: Experimental protocol. Main-paper figures report outcome comparisons; replication counts are mixed by experiment role and specified in Table 2.

Experiment role	Grid or comparison rule	Replication
CIFAR full-participation sweeps	$\lambda \in \{0, 10^{-3}, 0.005, 10^{-2}, 0.03, 0.05, 0.08, 0.1, 0.2, 0.5, 1, 2, 3\}$	one seed
CIFAR partial-participation sweeps	$\lambda \in \{0, 10^{-3}, 10^{-2}, 0.1, 0.5, 1\}$	one seed
Shakespeare fixed-Ditto sweeps	$\lambda \in \{0, 0.1, 0.5, 1\}$ at 10% and 50% participation	one seed
Tune-once shift tests	Select the full-participation fixed λ , freeze it, and reuse it under shifted participation without retuning	three seeds
CIFAR per-condition core comparisons	Re-sweep static Ditto for each dataset and participation level before comparing against AdaDitto	five seeds
Controller sensitivity scans	Vary AdaDitto controller configurations around the default: 28 configurations on CIFAR-100 practical and 11 configurations at each headline condition	three seeds for reported stress scans

Table 2: Sweep grids and replication rules. Single-seed sweeps identify coefficient regimes; multi-seed comparisons estimate robustness for the main claims.

Condition	λ^*	Mean gap	Gap > 0	Learned λ
CIFAR-path full	0.0	0.013	33%	0.090
CIFAR-path 20%	1.0	0.074	52%	0.135
CIFAR-prac full	0.5	0.008	30%	0.086
CIFAR-prac 20%	1.0	0.030	46%	0.101
Shakespeare 10%	> 0	0.003	5%	0.069

Table 3: Controller signal-validity audit. The CIFAR participation-shift rows show larger loss gaps and larger learned coefficients exactly where the best fixed λ moves upward.

Setting	Accuracy range	Spread
AdaDitto controller scan	0.4762–0.4791	0.0029

Table 4: CIFAR-100 practical controller scan over 28 configurations.

Quantity	CIFAR-path 20%	Shakespeare 10%
Seeds	3	3
Configs	11	11
Accuracy range	0.5627–0.5743	0.4819–0.4822
Spread	1.2 points	0.03 points
Default rank	6/11	5/11
Reference	tune-once 0.551	tuned 0.483
Outcome	all beat tune-once	all near tuned
Coeff. mean range	0.087–0.190	0.055–0.085

Table 5: Controller stress scans at the headline conditions. Both scans use three seeds per controller configuration and test whether the headline results depend on a lucky default controller setting.

Mode	Practical	Pathological
full controller	0.4689	0.5968
fixed base	0.4677	0.5976
no log ratio	0.4685	0.5971
no clipping	0.4689	0.5968
no EMA	0.4693	0.5971
mean reference loss	0.4681	0.5974
global shared coefficient	0.4657	0.5967
random adaptive	0.4768	0.5863

Table 6: Mean personalized accuracy for controller ablations over three seeds. The random-adaptive row is a negative-control sanity check; its practical-split value is not interpreted because all practical modes are close to seed variation, while it clearly hurts under high heterogeneity.

Tuned source	λ	1.0	0.5	0.2
Practical targets				
prac/1.0, prac/0.5	0.5	0.000	0.000	0.003
prac/0.2, path/0.2	1.0	0.002	0.000	0.000
path/1.0, path/0.5	0.0	0.025	0.033	0.072
AdaDitto, one config	–	0.017	0.021	0.032
Pathological targets				
prac/1.0, prac/0.5	0.5	0.022	0.008	0.003
prac/0.2, path/0.2	1.0	0.026	0.010	0.000
path/1.0, path/0.5	0.0	0.000	0.000	0.047
AdaDitto, one config	–	0.024	0.017	0.023

Table 7: Tune-once regret on CIFAR-100. Columns show target participation. Each cell reports accuracy lost relative to the target condition’s best fixed λ on the common grid $\{0, 0.001, 0.01, 0.1, 0.5, 1.0\}$. Fixed- λ sweep cells are single-seed diagnostics, so small differences should be interpreted cautiously; the large 4.7–7.2 point failures are the relevant signal.

Regime	Outcome
E=1, full participation	small or zero μ best
E=5, full participation	$\mu = 0$: 0.635; $\mu = 1.0$: 0.571
E=5, 20% participation	$\mu = 0$: 0.230; $\mu > 0$: 0.196
E=5, 20% with systems heterogeneity	$\mu = 0$: 0.259; $\mu = 1.0$: 0.222

Table 8: FedProx μ probes on CIFAR-100. Larger proximal penalties did not beat FedAvg-style $\mu = 0$ even under more local work and systems heterogeneity.

Fixed λ	10% participation	50% participation
0.0	0.344	0.445
0.1	0.483	0.512
0.5	0.483	0.511
1.0	0.483	0.511

Table 9: Shakespeare fixed-Ditto sweep. Nonzero coupling is essential, and the best coefficient remains stable across the tested participation levels.

Participation	Tuned Ditto	AdaDitto	Gap
10%	0.4822 ± 0.0028	0.4820 ± 0.0029	-0.0001
50%	0.5122	0.5118	-0.0004

Table 10: Shakespeare core comparison. AdaDitto matches tuned Ditto, but there is no shift win because the optimal fixed coefficient does not move.

Item	Details
Datasets	CIFAR-100 (Krizhevsky and Hinton 2009) and Shakespeare from the LEAF federated benchmark (Caldas et al. 2018). We do not introduce new datasets.
Data availability	CIFAR-100 is obtained from the public CIFAR distribution. Shakespeare is obtained through the public LEAF benchmark preprocessing pipeline. Raw datasets are not redistributed with this submission; users should follow the terms of the upstream sources.
Code availability	Experiments are implemented as modifications to PFLlib (Zhang et al. 2025). An anonymized code archive is provided at the URL in the main paper. Upon de-anonymization, code and experiment scripts will be released under the Apache-2.0 license inherited from the PFLlib codebase.
Hardware	Run manifests for the reported phases record an NVIDIA GeForce RTX 5060 Laptop GPU. The inspection environment reports an Intel Core Ultra 7 255HX CPU exposed to WSL2 as 8 logical CPUs and 23 GiB RAM. GPU VRAM and Windows host build were not captured in the run manifests; the WSL environment is Ubuntu 24.04.4 LTS on Linux 6.6.87.2-microsoft-standard-WSL2.
Software	Run manifests for the reported phases record Python 3.10.20, PyTorch 2.8.0+cu128, and CUDA 12.8. The active pflib environment used for artifact inspection contains torchvision 0.23.0+cu128, NumPy 2.2.6, pandas 2.3.3, SciPy 1.15.3, h5py 3.16.0, matplotlib 3.10.8, and cuDNN 9.10.2 as reported by PyTorch.
Randomness and seeds	Five-seed CIFAR core comparisons use seeds 0, 1, 2, 3, 4. Three-seed Shakespeare core comparisons, tune-once shift runs, and controller stress scans use seeds 0, 1, 2. Single-seed sweep-only configurations use seed 0. Each run applies the seed to Python, NumPy, PyTorch CPU, and PyTorch CUDA RNGs before data partitioning, model initialization, client sampling, and minibatch shuffling; client constructors additionally seed PyTorch with <code>seed + client_id</code> . Paired method comparisons use the same seed set.
Model and training hyperparameters	Final training hyperparameters are listed in Table 12. Controller hyperparameters use the default AdaDitto setting unless a sensitivity scan explicitly varies them: $\lambda_{\text{base}} = 0.08$, $\lambda_{\text{min}} = 0.05$, $\lambda_{\text{max}} = 3.0$, $\alpha = 0.8$, $\tau = 0.7$, $\gamma = 0.3$, $\beta = 0.9$, $\epsilon = 10^{-8}$, two warmup rounds, and a 10% probe sample capped at 256 examples.
Statistical reporting	No null-hypothesis significance tests are used for the claims in the main paper. Multi-seed core comparisons report mean and standard deviation across seeds and preserve paired seed comparisons whenever methods share the same seed set. Sweep-only configurations run with a single seed are used for hyperparameter selection and sensitivity visualization, not for significance claims.

Table 11: Reproducibility details for the experimental protocol. Values are taken from experiment scripts, run manifests, and the local environment where the artifacts were inspected; unavailable hardware fields are stated explicitly rather than inferred.

Dataset	Model	Rounds	Clients / participation	Local epochs	Batch size	Optimizer / LR
CIFAR-100 practical	CNN	200	20 / full, 50%, 20%	1	64	SGD / 0.01
CIFAR-100 pathological	CNN	200	20 / full, 50%, 20%	1	64	SGD / 0.01
Shakespeare 10%	char-LSTM	40	118 / 10%	1	64	SGD / 0.5
Shakespeare 50%	char-LSTM	40	118 / 50%	1	64	SGD / 0.5

Table 12: Final training hyperparameters used in the reported experiments. All rows use SGD with no weight decay. Ditto and AdaDitto train both the global and personalized models with the same local learning rate; the personalized Ditto optimizer adds the proximal step with the fixed or adaptive coefficient.

References

- Caldas, S.; Duddu, S. M. K.; Wu, P.; Li, T.; Konečný, J.; McMahan, H. B.; Smith, V.; and Talwalkar, A. 2018. LEAF: A Benchmark for Federated Settings. ArXiv:1812.01097.
- Krizhevsky, A.; and Hinton, G. 2009. Learning Multiple Layers of Features from Tiny Images. Technical report, University of Toronto.
- Zhang, J.; Liu, Y.; Hua, Y.; Wang, H.; Song, T.; Xue, Z.; Ma, R.; and Cao, J. 2025. PFLlib: A Beginner-Friendly and Comprehensive Personalized Federated Learning Library and Benchmark. *Journal of Machine Learning Research*, 26(50): 1–10.